SRI BALAJI YELLELA

Fullstack Software Engineer

+1 (402) 500-0033 | yellelasribalaji@gmail.com | linkedin.com/in/sriyellela | github.com/Sri-Yellela | Boston, MA

TECHNICAL SKILLS

- **Languages Go** · Java · Python · C# · TypeScript · JavaScript · **SQL**
- **Backend & Distributed Distributed systems** · **Microservices** · REST APIs · gRPC · Spring Boot · Node.js · .NET · Event streaming
- **Data Stores PostgreSQL** · DynamoDB · SQL Server · Azure SQL Database · CosmosDB · MySQL
- **Infrastructure Kubernetes** · Docker · Terraform · CI/CD pipelines · AWS · GCP · Microsoft Azure
- **Testing** JUnit · Mockito · Jest · Cypress · Selenium · Integration testing
- **Practices** Agile/Scrum · Test-driven development · Object-oriented design · Code review
- **Transferable to target stack** GCP Pub/Sub and AWS SQS (durable message brokers, equivalent role to **Kafka**) · EKS (managed **Kubernetes**)

EXPERIENCE

Stripe | Payments Infrastructure, Bangalore — Aug 2022 - Dec 2023

Software Development Engineer

- Engineered payment authorisation services in **Python** and **Go** that sustained **1M+ daily transactions** at **99.9% uptime**, keeping settlement accurate during peak merchant load.
- Partitioned a monolithic reconciliation job into **microservices** coordinated through **GCP Pub/Sub**, clearing message backlogs that had been breaking settlement for **12 downstream services**.
- Tuned **REST API** request paths and connection pooling to hold **sub-100ms p99 latency** across **200K+ merchant integrations**, which kept checkout responsive as volume grew.
- Chose at-least-once delivery with idempotency keys over exactly-once semantics after weighing operational cost, accepting duplicate handling in exchange for a simpler failure mode the on-call team could reason about.
- Instrumented tracing and alerting across critical payment paths, cutting **mean time to recovery by 35%** and giving merchant support a shared signal to triage against.

Amazon | Supply Chain & Internal Developer Tools, Hyderabad — Jan 2021 - Jul 2022

Software Development Engineer

- Operated cloud-native services on **AWS** (**Lambda**, **SQS**, S3, **EKS**) serving **500K+ daily queries** at **98% availability** for warehouse planners.
- Diagnosed and resolved **PostgreSQL** and **DynamoDB** query bottlenecks that were stalling **150K+ daily inventory operations**, restoring same-day stock accuracy.
- Shipped weekly zero-downtime rolling updates on **Kubernetes**, replacing a maintenance-window release that had blocked fulfilment during deploys.
- Refactored core ordering modules onto consistent **object-oriented design** patterns, reducing module-level defect rates by approximately **30%** and shortening review cycles.
- Presented service design proposals at technical review and guided incoming engineers through the supply chain domain, which shortened their first-commit time.

ACADEMIC PROJECTS

Distributed Key-Value Store with Consensus Protocol | Graduate coursework — Spring 2024

Java, Raft consensus, gRPC, Docker, JUnit

- Implemented a **distributed** key-value store on the **Raft consensus** algorithm with leader election and log replication, holding linearisable reads across a 5-node cluster under simulated network partitions.
- Measured **12K ops/sec** read throughput under load with concurrent failure injection, validating the replication design against the course fault model.

Cloud-Native ETL Pipeline on GCP | Graduate coursework — Fall 2024

Python, Apache Beam, GCP Dataflow, BigQuery, Pub/Sub, Terraform

- Delivered a streaming ETL pipeline moving **100M+ records** from **Pub/Sub** into **BigQuery** with schema validation and failure alerting, holding data loss under **0.1%**.
- Codified the pipeline environment in **Terraform**, cutting full redeployment from 45 minutes to under 4 minutes for each grading iteration.

PROJECTS

Full-Stack Scalable Web Platform — 2024 - 2025

React, TypeScript, Spring Boot, PostgreSQL, Docker, JWT, OpenAPI

- Built a **REST API** backend in **Spring Boot** over **PostgreSQL** with a React and **TypeScript** client, holding sub-200ms average response for 200 concurrent users.
- Covered the service layer with **JUnit**, Mockito and RestAssured suites, catching contract regressions before they reached the deployed environment.

EDUCATION

Northeastern University | Boston, MA — Jan 2024 - Dec 2025

Master of Science, Computer Science